BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Merge Sort

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What does insertion sort do with the next unsorted item?

2. Which task suits utility software?

A) Writing an essay
B) Computer maintenance
C) Editing a photograph
D) Playing a game

3. "File management is only the execution of arithmetic instructions." Correct this. Explain why.

4. A teacher stores private records on a portable drive. "An encryption utility makes a drive weigh less." Correct this. What does it protect?

Today's objectives

Show recursive splitting into smaller parts conceptually. Explain the merge of two sorted lists.

Key vocabulary - Read, cover, recall

Merge sort
Divide-and-conquer sorting method that merges ordered sublists.

Divide and conquer
Split a problem into smaller subproblems, then combine solutions.

Split
Separate the list into smaller parts.

Merge
Combine sorted lists into one ordered list.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Divide the problem	Merge sort divides a collection into halves until single elements remain.
Single elements are sorted	A list containing one item needs no further sorting.
Merge in order	Merge two sorted halves by comparing their front values, taking the smaller one and repeating.
Rebuild the whole list	The smaller sorted lists combine step by step to produce one completely sorted result. OCR requires the main steps, not code memorisation.

I Do - One worked example

Trace merge sort on [8,3,5,1]. Show the split, the sorted pairs and the final sorted list. [4 marks]

1. Split into [8,3] and [5,1] [1].

2. Split into individual elements [8],[3],[5],[1] [1].

3. Merge into sorted pairs [3,8] and [1,5] [1].

4. Merge these pairs in order: [1,3,5,8] [1].

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Merge sort [6,2,7,1]. Give the sorted pairs and final result. [4 marks]

2. Explain why merge sort splits [9,3,4,2] down to single values. [2 marks]

3. Merge the sorted lists [2,7] and [3,6] into one sorted result. [4 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition choosing the smaller front value while merging two sorted lists.

```
array left = [3,8]
array right = [1,5]
if ... then
print(left[0])
else
print(right[0])
endif
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the key idea behind the divide step of merge sort. [2 marks]

Q3. Trace merge sort on [4,1,3,2], showing two sorted pairs and the final result. [4 marks]

Q4. Explain the merge-sort process for [7,2,6,1], including split, comparison and merge. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. Merge sort combines sorted smaller lists to produce a sorted full list. ☐ T ☐ F
2. Merge sort divides a collection into halves until single elements remain. ☐ T ☐ F
3. A list containing one item needs no further sorting. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Place the next item in the correct position in the sorted part. Shift larger items right. ☐

Q2 B - Computer maintenance ☐

Q3 It manages files and folders: naming, saving, copying, moving and deleting them. ☐

Q4 Encryption uses an algorithm and key to change readable data into ciphertext. It protects privacy; compression reduces size and defragmentation rearranges HDD data. ☐

Code First - Model

```
array left = [3,8]
array right = [1,5]
if left[0] < right[0] then
print(left[0])
else
print(right[0])
endif
```

Output: 1

We Do - Check the model

1. Split [6,2] and [7,1] [1]. Sort pairs as [2,6] and [1,7] [1]. Compare leading elements when merging [1]. Final order [1,2,6,7] [1]. ☐

2. Single values are already sorted [1] and can then be merged into correctly ordered larger groups [1]. ☐

3. Compare 2 and 3, take 2 [1]. Compare 7 and 3, take 3 [1]. Compare 7 and 6, take 6 [1]. Then append 7 -> [2,3,6,7] [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Split an array into smaller halves [1] until individual items/single-element lists remain [1]. ☐

Q3 Split into [4,1] and [3,2] [1]; merge singletons into [1,4] [1] and [2,3] [1]; final merge [1,2,3,4] [1]. ☐

Q4 Split into [7,2] and [6,1] [1]; split into singletons [1]. Merge into [2,7] [1] and [1,6] [1]. Compare leading values of sorted halves and take smaller repeatedly [1], producing [1,2,6,7] [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: